

India's 3-STAGE NUCLEAR POWER PROGRAMME

From Natural Uranium to a
Sustainable Nuclear Future



Stage I
PHWRs



Stage II
FBRs



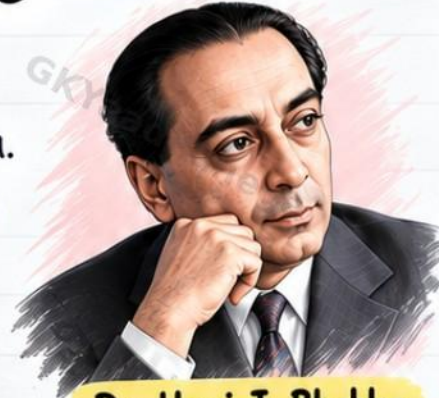
Stage III
Thorium-based



India's Three-Stage Nuclear Power Programme

What is it?

- A long-term nuclear energy strategy for India.
- Proposed by **Dr. Homi J. Bhabha** in the 1950s.
- Aims to achieve maximum utilisation of India's **thorium resources** through a three-stage approach.



Dr. Homi J. Bhabha

Architect of India's nuclear programme

Why was it needed?

- Limited domestic **uranium reserves**.
- India has **large thorium reserves**.
- Need for **long-term and sustainable** energy source.
- To **reduce dependence** on imported fossil fuels.
- To utilise indigenous resources for **energy security and self-reliance**.



Uranium
(limited reserves)



Thorium
(large reserves)

The Three-Stage Approach at a Glance

Stage I

Natural Uranium

↓
PHWRs

(Pressurised Heavy Water Reactors)

Generates electricity and produces **Plutonium-239**



Stage II

Plutonium-239 + Uranium

↓
Fast Breeder Reactors (FBRs)

Breeds more fissile material and produces **U-233**



Stage III

Thorium-232

↓
Thorium-based Reactors
(e.g. AHWR)

Produces **U-233** and generates large-scale electricity



Stage I

Pressurised Heavy Water Reactors (PHWRs)

- **Main fuel** : Natural uranium (mostly U-238).
- **Moderator** : Heavy water (D_2O).
- **Coolant** : Heavy water (D_2O).
- **Type** : Thermal reactors using low enriched/natural uranium.
- **Function** : Generate electricity and produce **plutonium-239 (Pu-239)**



PHWR

(Pressurised Heavy Water Reactor)

Why PHWRs were chosen for India?

- Can use **natural uranium** (no enrichment required).
- Suitable for India's **limited uranium reserves**.
- Heavy water technology **developed indigenously**.
- Provides plutonium-239 for Stage II.
- Suited to India's long-term strategy.



Natural Uranium Fuel Rods
(used in PHWRs)

Key Terms

- **Natural Uranium** : Contains ~0.7% U-235 and ~99.3% U-238.
- **U-238** : Fertile material (not fissile).
- **U-235** : Fissile material (supports fission).
- **Plutonium-239 (Pu-239)** : Fissile material produced in PHWRs, used as fuel in Stage II.

Stage II

Fast Breeder Reactors (FBRs)

- **Main fuel** : Plutonium-239 (from Stage I) + Uranium (U-238).
- **Type** : Fast neutron reactors.
- **Coolant** : Liquid sodium (not heavy water).
- **Moderator** : None (fast neutrons used).
- **Function** : Breeds more fissile material and produces Uranium-233 (U-233).
- **Role** : Bridges Stage I and Stage III.



Prototype Fast Breeder Reactor (PFBR)

Kalpakkam, Tamil Nadu

How does FBR work?

- 1 Uses Plutonium-239 as fuel.
- 2 Fast neutrons convert U-238 into more Pu-239 (breeding process).
- 3 Also helps in producing U-233.
- 4 Thus, it multiplies India's limited uranium resources.



Mixed Oxide (MOX) Fuel Rods
(Pu-239 + U-238)

Importance for India

- Increases availability of fissile material.
- Makes more efficient use of India's limited uranium.
- Provides fuel for Stage III (U-233).
- Key step towards energy self-reliance.

Stage III

Thorium-Based Reactors

- **Main fuel** : Thorium-232 (Th-232)
- Thorium is a **fertile material** (not fissile).
- Converted into **Uranium-233** (U-233) through neutron capture and decay.
- **Type** : Advanced Heavy Water Reactors (AHWR) - proposed design.
- **Function** : Generates electricity using **Uranium-233** produced from thorium.



Advanced Heavy Water Reactor (AHWR)
(Thorium-based)

Why Thorium is Important for India?

- 1 India has **large thorium reserves** (3rd largest in the world).
- 2 Ensures **long-term energy security**.
- 3 Reduces dependence on imported uranium.
- 4 Provides **sustainable and indigenous source of nuclear energy**.



Thorium-232
Fuel Form
(example)

Key Terms

- **Thorium-232 (Th-232)** : Fertile material (not fissile).
- **Uranium-233 (U-233)** : Fissile material produced from Th-232.
- **AHWR** : Proposed indigenous reactor for Stage III.

How the Three Stages are Linked?

1 Stage I (PHWRs)

- Uses natural uranium.
- Produces **Plutonium-239** (Pu-239)



PHWR

2 Stage II (FBRs)

- Uses Pu-239 (from Stage I) + Uranium.
- Produces **Uranium-233** (U-233).



FBR

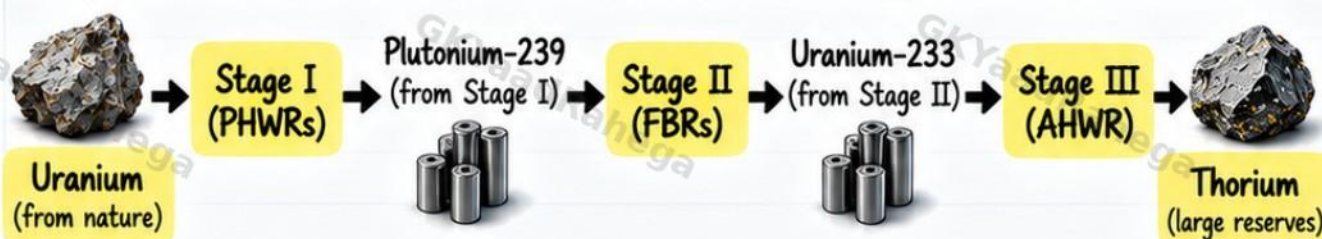
3 Stage III (Thorium-based)

- Uses **Thorium-232** + U-233 (from Stage II).
- Generates large-scale electricity.



AHWR
(Thorium-based)

Flow of Fuel in India's Three-Stage Programme



Role of Thorium in India's Energy Security

- India has one of the **largest thorium reserves** in the world.
- Helps in **long-term** and **sustainable energy supply**.
- **Reduces dependence** on imported uranium.
- Supports India's goal of **energy self-reliance**.

Key Advantages of the Three-Stage Programme

- 1 Maximum utilisation of India's thorium resources.
- 2 Ensures long-term energy security.
- 3 Reduces dependence on imported fossil fuels.
- 4 Utilises indigenous technology and resources.
- 5 Provides sustainable and clean energy source.
- 6 Supports India's goal of energy self-reliance.



Thorium
(large reserves)



Clean & Sustainable
Energy

Challenges in Implementation

- 1 Advanced technology required (FBRs and AHWRs).
- 2 High initial cost of reactors.
- 3 Long gestation period for Stage II and III.
- 4 Need for continuous R&D and technological development.
- 5 Safety and waste management challenges.



Advanced Nuclear
Technology
(FBR / AHWR)

Significance for India

- 1 India has limited uranium but large thorium reserves.
- 2 The programme converts this challenge into an advantage.
- 3 Ensures long-term, sustainable and self-reliant energy future.
- 4 Helps meet growing electricity demand.
- 5 Positions India as a global leader in thorium-based nuclear technology.



Energy
Self-Reliant
India



**COMPLETE
NOTES
BUNDLE**

- ✓ Concept Clarity
- ✓ Exam Oriented
- ✓ Easy Language
- ✓ One Time Purchase



COMPLETE NOTES BUNDLE

All 11 Subject Notes in 1 Bundle

Complete

Exam-Focused

Latest Pattern

One Time Purchase

History



- Ancient History
- Medieval History
- Modern History

Art & Culture



- Indian Art & Culture
- Dance, Music
- Architecture
- Literature

Geography



- Indian Geography
- World Geography
- Physical Geography

Polity



- Constitution
- Governance
- Parliament
- Judiciary

Economics



- Basic Concepts
- Indian Economy
- Banking
- Budget

General Science

(Physics + Chemistry + Biology)



Environment & Ecology



- Ecology
- Biodiversity
- Environment
- Climate Change

Computer



- Fundamentals
- Applications
- Internet & Networking

Static GK



- Awards
- Sports
- Books
- Important Days
- Miscellaneous GK

Science & Technology



- Space
- Defence
- Emerging Technologies
- Major Missions

Important Theme Current Affairs



- Exam-focused Themes
- National & International
- Important Developments



ONLY AT

₹499/-

**BEST
VALUE**

**11 Subjects
Complete Notes
One Time Purchase
at Just ₹499/-**



Well
Structured



Exam
Focused



Latest
Pattern



Concept
Clarity

**Join Our Telegram Channel
for Sample Pages & to Purchase PDF**



JOIN NOW

@GKYaadRahega